04 CONTROL FLOW • THEORY

01 conditional if else and switch

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

01 — Conditional Selection: `if-else` and `switch-case`

1. `if-else` Pitfall: Assignment `=` vs. Equality `==`

Writing `if (x = 5)` assigns `5` to `x` and evaluates to `true` (because `5 ≠ 0`)! Always use `==` for comparison.

2. `switch` Statement Rules (`[stmt.switch]`)

- The condition of a `switch(expr)` **must** have an **integral or enumeration type** (`int` , `char` , `enum`). Floating-point types (`float` , `double`) and `std::string` are **Compilation Errors** in `switch` !
- Every `case` label must be an integral constant expression (`constexpr`).
- Omitting `break;` causes execution to **fall through** into the next `case` block!